TEAM: ir gonna win

Albertaloop Mechatronics Showdown 2026

Autonomous Navigation · Balance Beam · Design Evaluation

Keiron Street · Param Vyas · Kumaran Vaitheeswaran · Aahil Ansarr · Alexander Macaloney

Group 10

Challenge Overview

15 pts

Balance Beam

Autonomous balancing using IMU + PID control

20 pts

Manual Maze

Controller navigation through maze checkpoints

30 pts

Autonomous Maze

Left-wall following + IR collision detection

60 pts

Design Eval

Presentation, process & engineering rationale

Total: 125 Points Available

Team & Role Assignments

Electrical / Electronics

Kumaran Vaitheeswaran · Keiron Street

- Wired schematic using colour-matched jumper wires for traceability
- Integrated ultrasonic and IR sensors via breadboard power rails
- Keiron simulated gyro sensor virtually on Wokwi to save iteration time
- Keiron developed PID balance beam logic with bubble meter calibration

Mechanical

Param Vyas · Alexander Macaloney

- Built chassis — wheels, actuators, base plate
- Tested all 4 motors for uniformity before integration
- Screwed motors inward for compact design
- Mounted 6V batteries at mid-section bottom for lower COG and stability
- Created ultrasonic sensor mount attachment for front of robot

Software / Integration

Aahil Ansarr

- Primary coder — wrote and iterated all Arduino sketches
- Re-wired full schematic onto chassis for clean integration
- Used chassis holes to route wires internally, reducing exposed wiring
- Integrated electronics with chassis and drivetrain

Documentation

Kumaran Vaitheeswaran

- Wrote technical report and updated progress throughout build

Process Workflow — Day 1

Mechanical

Chassis build & motor test

Built chassis with wheels and actuators. Tested all 4 motors for uniformity. Created test base with two battery packs — one per side for independent left/right power control to enable dynamic turning.

Electrical

Schematic wiring + sensor integration

Reproduced starter code schematic using colour-matched wires for traceability. Extended onto breadboard with makeshift 5V/GND rails to integrate ultrasonic and IR sensors.

Observation

IR sensor thermal radiation discovery

IR sensor triggered at ~4cm with a human finger but at a different distance with wood. Root cause: human skin emits thermal radiation that IR receivers respond to differently than non-organic materials. Potentiometer tuning needed per surface.

Software

Wokwi virtual simulation

Keiron simulated gyro sensor on Wokwi using matching virtual modules, saving physical iteration time on IMU wiring and logic design.

Issue

Uneven motor voltage — end of Day 1

Multimeter confirmed different voltages reaching all 4 wheels. Motors refused to spin consistently. Root cause not yet identified. External help sought.

Process Workflow — Day 2

Mechanical

Full rewire with fresh batteries

Param rewired entire vehicle from scratch. New batteries in parallel — all motors spinning, though unevenly. Loose wires taped by Alex during testing. Three motors confirmed operational immediately.

Electrical

Arduino power fix — 9V direct supply

Observed that 5V from L298N was powering the Arduino — insufficient and unstable. Disconnected and powered Arduino directly via 9V external source. Power stability improved.

Mechanical

Batteries: parallel → series

Parallel wiring delivered only 4.6V combined from 6V packs — too low for reliable motor operation. Switched to series configuration to deliver full voltage to motor drivers.

Software

Turn logic improvement

Original code turned by spinning one wheel forward and the opposing wheel backward. Modified so both wheels on one side spin together — vastly improved turn consistency and radius.

Software

Balance beam: bubble meter calibration

Measured max angular displacement of balance beam using a bubble meter. Used known tilt range to set PID baseline and maximum allowed tilt, calibrating the robot to the specific beam geometry.

Hardware & System Architecture

Controller	Arduino Mega 2560
Drive	4WD Differential — 4× DC Motors
Motor Driver	2× L298N H-Bridge
Power	2× 6V AA Battery Packs
Right Wall	HC-SR04 Ultrasonic (Pins 22/23)
Front Detect	Digital IR Sensor (Pin 24)
Right Detect	Digital IR Sensor (Pin 25)
Balance	GY-521 MPU6050 IMU (I2C 20/21)

Pin Assignments

```
Pins 2–13   →  Motor control (EN, IN1, IN2)
Pin 22      →  Ultrasonic TRIG
Pin 23      →  Ultrasonic ECHO
Pin 24      →  Front IR OUT
Pin 25      →  Right IR OUT
Pin 20      →  IMU SDA (I2C)
Pin 21      →  IMU SCL (I2C)
A0 / A1     →  Joystick X / Y
```

Autonomous Maze Navigation — Final V4.0

Right-Hand Wall Following + Gyro Turns + Backup Clearance

Library: MPU6050_light — uses mpu.calcOffsets() + mpu.getAngleZ() for absolute heading

P1

Front IR triggered

Stop → backup 450ms → turnLeftGyro() to 88° via mpu.getAngleZ()

P2

Right wall < 4cm or IR high

steerLeft() — slow left side to 100 PWM, keep right at 255

P3

Right wall 4-12cm

moveForward(220) — wall perfectly in range, drive straight

P4

Right wall > 12cm or 999

steerRight() — slow right side, veer back toward wall

`BASE_SPEED 220 · TURN_SPEED 255 · STEER 255:100 · WALL_TOO_CLOSE 4cm · WALL_TOO_FAR 12cm · TURN_TARGET 88° · BACKUP 450ms · MIN_PWM 72`

Balance Beam — Autonomous Control

1
Approach

Drive forward
at APPROACH_SPEED
(120 PWM)

2
Detect

Pitch crosses
±2° threshold
→ switch to HOLD

3
Hold

PID corrects
pitch toward
−2.5° target

4
Success

Hold ≤2° error
for 5 consecutive
seconds

PID + Complementary Filter

Kp = 8.0 — Proportional: corrects proportionally to how far pitch deviates from −2.5°

Ki = 0.05 — Integral: accumulates over time to eliminate long-term drift (kept small to prevent windup)

Kd = 1.2 — Derivative: dampens overshoot before it happens

Filter: 98% gyroscope (smooth) + 2% accelerometer (absolute reference) = stable pitch angle

Engineering Design Process

Left-hand wall following

Rationale: Guarantees maze completion for any simply-connected maze. Simpler than flood-fill, more reliable than random turn.

Result: Consistent corridor navigation with single ultrasonic sensor

Digital IR over analog distance

Rationale: HC-SR501 PIR in kit detects heat, not walls. Digital IR obstacle sensors confirmed allowed by organizers. Potentiometer allows physical distance calibration without code changes.

Result: Tunable front + right detection at competition

Complementary filter over library

Rationale: Raw I2C register access gives full control over sensor fusion. 98/2 gyro/accel blend eliminates drift while suppressing vibration noise from motors.

Result: Stable pitch angle despite motor vibration on beam

PWM minimum threshold of 72

Rationale: Motors buzz but don't move below this value on ECERF carpet. Discovered during motor testing — prevents wasted power and false motion signals.

Result: Reliable motor response on competition surface

Motor Troubleshooting

1 Motors refused to spin

Near end of Day 1 — all motors suddenly stopped responding despite code running correctly.

2 Physical wiring check

Inspected all connections — wiring appeared intact. No loose connectors found.

3 Multimeter diagnosis

Measured voltage entering and leaving both L298N drivers. Found a 1.4V drop between input and motor output.

4 Voltage drop identified as normal

Researched L298N specs — 1.4V internal drop is normal H-bridge operation, not a fault. Initial diagnosis was a red herring.

5 Full rewire on Day 2

Rewired entire vehicle from scratch. Three motors immediately operational.

6 Fourth motor — loose ENB wire

Identified a loose connection on the ENB pin of one L298N driver. Correcting this brought the fourth motor online.

Other Challenges & Solutions

Wiring

Motor direction inconsistency across chassis

Motors physically mounted in opposing orientations on left vs right side. Resolved by individually testing each motor and mapping confirmed HIGH/LOW directions — not assuming symmetry.

Code

steerLeft had no visible effect on carpet

BASE_SPEED / 2 = 50 PWM, below the 72 minimum threshold. Hardcoded steer values to 150/80 — both sides now clear the friction threshold.

Sensors

IR sensor misidentified as distance sensor

HC-SR501 in kit is a PIR (heat sensor), not a distance sensor. Sourced confirmed-allowed digital IR obstacle sensors. Detection distance set via onboard potentiometer.

Code

Fixed TURN_TIME unreliable on carpet

Replaced time-based turning with gyro integration — robot turns until MPU6050 measures 85° of yaw. Consistent regardless of battery charge or carpet friction.

Code Highlights

Gyro turn — MPU6050_light

```
mpu.update();
float start = mpu.getAngleZ();

setTurnLeftDirections();
setSpeed(TURN_SPEED, ...);

while (abs(mpu.getAngleZ()
       - start) < 88.0) {
  mpu.update();
}
```

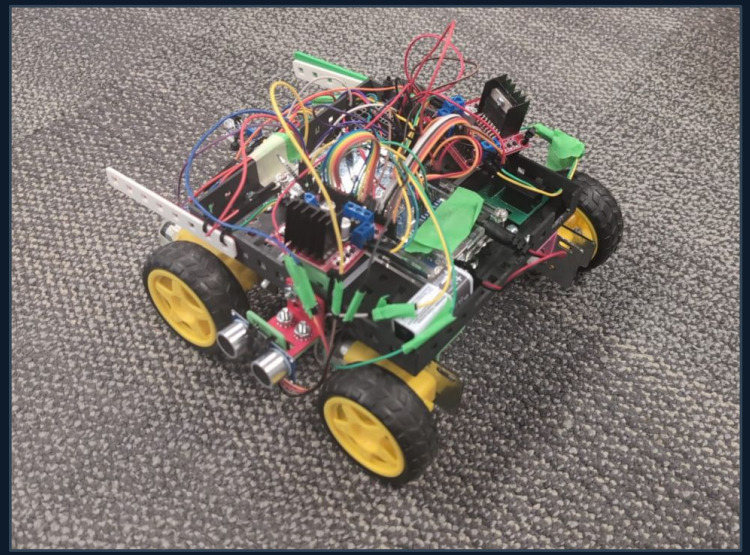
mpu.getAngleZ() returns absolute heading. Absolute delta vs time-based integration — no drift, consistent on any surface. calcOffsets() zeros the sensor on startup.

Right-wall follow loop

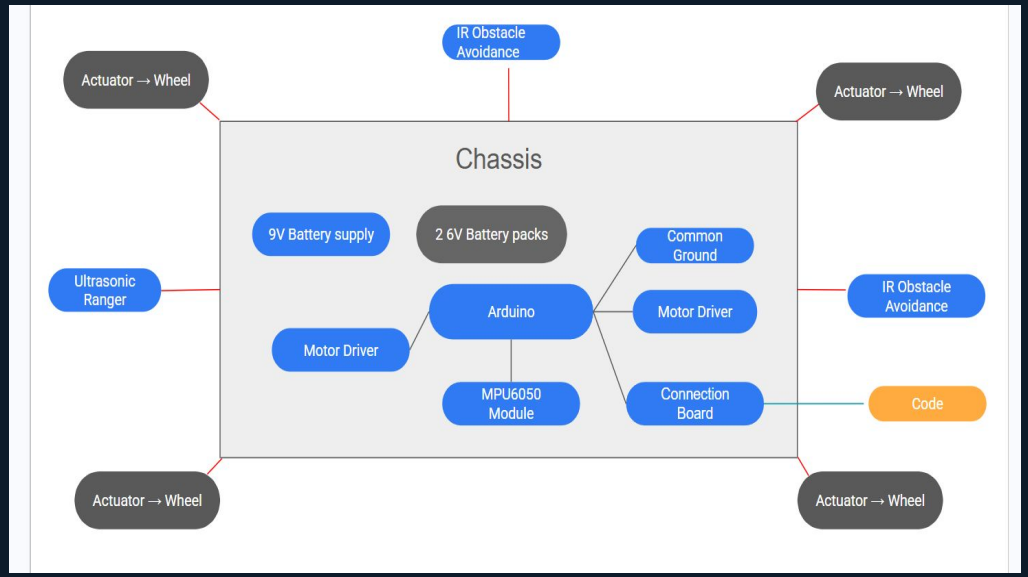
```
mpu.update(); // every loop

if (frontWall) {
  backup → turnLeftGyro();
}
if (rightDist < 4 || rightIR) {
  steerLeft();  // too close
}
if (rightDist >= 4 &&
    rightDist <= 12) {
  moveForward(220); // in range
}
if (rightDist > 12) {
  steerRight(); // find wall
}
```

Ultrasonic on right side. IR provides binary confirmation when wall is very close. Returns 999 (not 0) on timeout — avoids false 'wall lost' triggers.

Robot Build



Full robot — top view



Full Mechatronic Schematic

Team Keiron

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3	4WD	3	PID
Challenges attempted	Differential drive	Sensors integrated	Balance algorithm

Key Takeaways

— Sensor selection matters — confirming sensor type before integration saved significant debugging time

— Physical testing revealed motor direction and PWM threshold issues not visible in code alone

— Complementary filter provided stable IMU readings despite motor vibration interference

— Left-hand wall following is provably complete for the given maze topology